

The source code of this project is available for free download at source.efymag.com

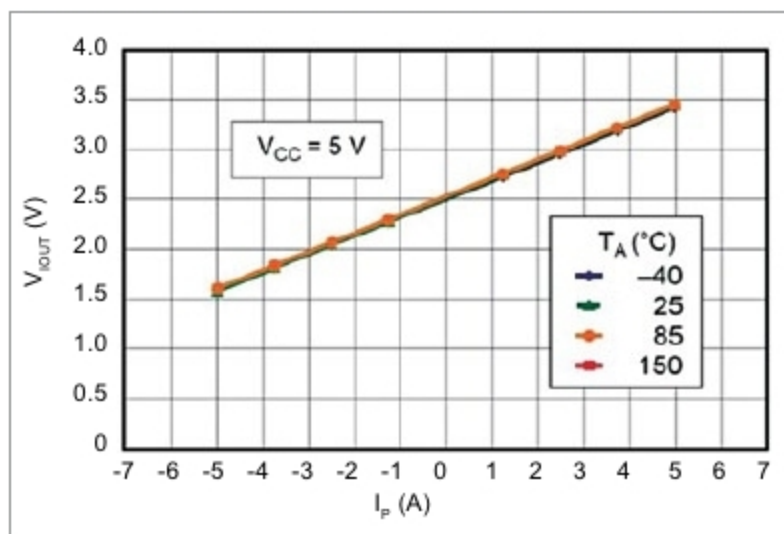


Fig. 3: Output voltage versus sensed current of ACS712

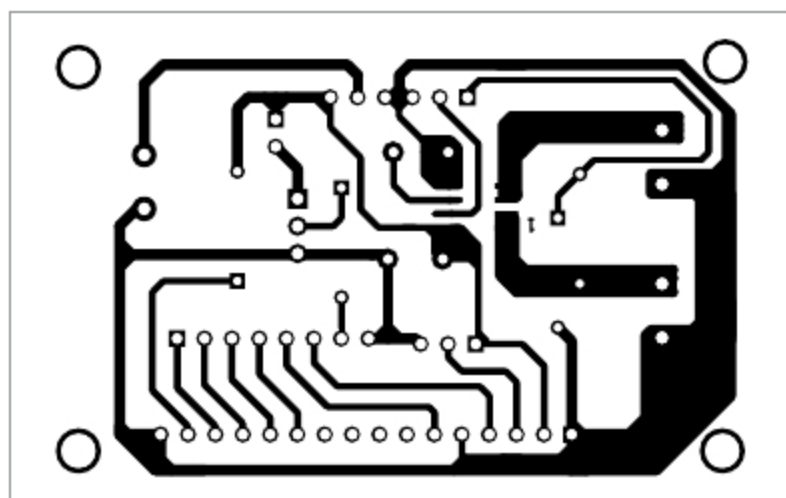


Fig. 4: Actual-size PCB layout of DC panel meter using Arduino

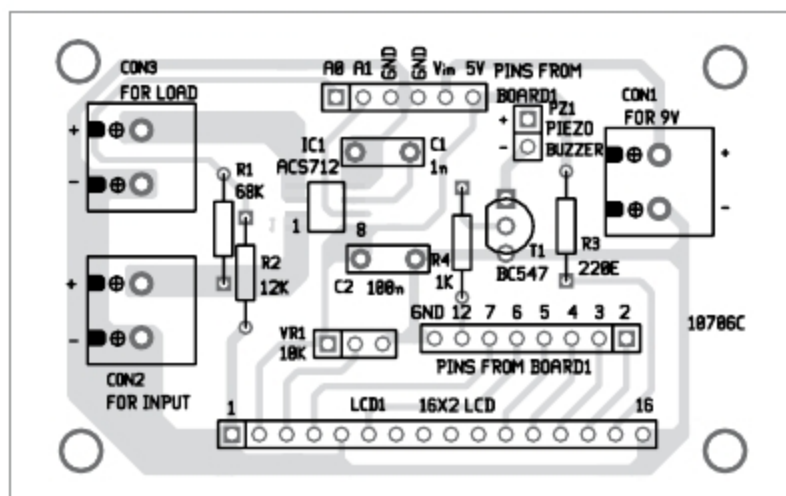


Fig. 5: Components layout for the PCB

Software is developed on Arduino platform. After burning the code (PanelMeter.ino) into ATmega328 on

connecting the circuit to Arduino board. Arduino can be powered by an external 9V, 500mA adaptor or USB

Arduino Uno, remove USB power supply from the computer/laptop. Connect 9V power supply across CON1.

During testing, 5V, 5A input was used as power input source at CON1. Load connected at CON3 may have different values (ranging from 1E to 5E). A 10W wirewound resistor is used as load across CON3. Current and voltage values appear on LCD1. When current flows more than 5A, LCD1 shows over range and the buzzer produces a sound to alert the user. A low-current, disk-type piezo-electric buzzer is used.

Construction and testing

An actual-size PCB layout of the DC panel meter using Arduino is shown in Fig. 4 and its components layout in Fig. 5. After assembling the circuit on the PCB, and burning the code to the MCU, connect a 9V across CON2 to power the circuit. Connect potentiometer VR1 on the front panel of the cabinet for controlling the contrast of LCD1. Connect 5V, 5A input across CON1 and load across CON3 for sensing the current.

Use jumper wires for

A-to-B cable.

If current is displayed with negative (-) sign, it either means that it is flowing in opposite direction or input pins of current sensor IC are interchanged. To use a high-current, coil-type buzzer, use a driver with external power supply instead of Arduino's supply. For additional protection, connect 4.7V Zener diode between analogue input pins (A0 and A1) and ground of Arduino. **EFY**



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